Formatting Instructions For NeurIPS 2026

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Abstract

Concept erasure in language models can degrade model utility. One plausible mechanism is that erasure directions computed in high-dimensional ambient space can point orthogonal to the data manifold, producing off-manifold edits that displace representations into low-density regions. We present Gradient-based Concept Erasure with Tangent projection (`gce_tangent`), which constrains erasure gradients to the local tangent space of the representation manifold. On Qwen2.5-0.5B, the completed paper runs on sycophancy, safety, and Bias in Bios consistently show lower measured leakage for `gce_tangent` than for `gce_ambient`. The current evidence is narrower than our original thesis, however: the lower-leakage tangent settings often incur larger displacement and mixed utility retention, so the completed results support stronger removal but not yet a uniformly better leakage–utility frontier. Geometry diagnostics still strengthen the mechanism story by showing that sycophancy is substantially more nonlinear than gender, making manifold-aware erasure a promising direction for surgical concept removal.

1 Introduction

Effective concept erasure in large language models requires a delicate balance between removing targeted information and preserving the model's broader utility. While gradient-based methods offer a path toward targeted intervention, they often suffer from substantial performance degradation. A natural hypothesis is that this collapse is driven in part by off-manifold edits: erasure directions computed in the high-dimensional ambient space can point orthogonal to the data manifold, displacing representations into low-density regions where the model's computation becomes unstable.

To address this failure mode, we propose Gradient-based Concept Erasure with Tangent projection (`gce_tangent`). By constraining erasure gradients to the local tangent space of the representation manifold, `gce_tangent` aims to remove concept-specific structure while preserving more of the underlying distributional geometry. This approach contrasts with `gce_ambient`, which applies raw gradients in the full hidden space and therefore provides a direct baseline for the value of tangent projection.

Our investigation focuses on sycophancy as the primary nonlinear concept, representing a challenging case for alignment-oriented concept removal. Bias in Bios (gender) serves as a diagnostic baseline, and an iterative safety setting tests whether the same geometric idea extends to a separate surgicality problem. Using Qwen2.5-0.5B as our experimental platform, we evaluate these methods across multiple layers where completed evidence is available; within the paper-locked safety suite, the completed comparison is currently centered on layer 8. The completed paper runs support a narrower conclusion than our original draft: `gce_tangent` consistently lowers measured leakage relative to ambient baselines, but the present evidence does not yet show that it also dominates the utility-displacement tradeoff. We therefore frame this manuscript as evidence for a removal advantage and for a strong geometry contrast between concepts, while leaving the stronger frontier claim to broader iterative and matched-leakage analyses.

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2 Method

We implement Gradient-based Concept Erasure (GCE) using an iterative protocol with probe recycling. Each round of erasure begins by computing per-sample gradients from a current multi-layer perceptron (MLP) scorer. These gradients are normalized to unit directions. In the tangent variant, the algorithm projects these directions onto the local tangent space of the representation manifold. The erasure step modifies the hidden state by subtracting the projection of the state onto the erasure direction, scaled by a step size parameter.

To estimate the local tangent space for each representation, we find its k-nearest neighbors in the current data. Secant vectors are computed between the target point and its neighbors. Running principal component analysis (PCA) on these secant vectors allows us to extract the top principal components, which form the tangent basis. The tangent estimator is refit in every round to track the evolving manifold geometry as representations shift.

Two diagnostics help evaluate tangent quality. Tangent Gradient Capture (TGC) measures the fraction of the leakage gradient that lies within the estimated tangent space. Tangent Neighbor Residual (TNR) measures the fraction of local secant vectors not explained by the tangent basis. These metrics verify that the tangent estimation remains geometrically sound throughout the iterative process.

3 Experiments

Our experimental evaluation focuses on characterizing the performance of `gce_tangent` against `gce_ambient` across three distinct concept domains. We prioritize nonlinear concepts where manifold geometry is expected to play a critical role, while maintaining a diagnostic baseline for linear comparison.

3.1 Experimental Domains and Ordering

We evaluate our methods across three primary blocks, ordered by their relevance to nonlinear manifold erasure:

1. **Sycophancy (Primary Nonlinear Case):** Our central focus is on removing agreement bias in model responses. This represents a critical challenge for model alignment where concept encoding is expected to be locally curved.
2. **Safety (Secondary Case Study):** We evaluate the surgicality of erasure on content safety using the PKU-SafeRLHF dataset, testing the preservation of helpfulness alongside safety leakage.
3. **Bias in Bios (Diagnostic Baseline):** We use gender and profession classification as a replication anchor. This setting provides a linear comparison point to contrast against the nonlinear gains of tangent-constrained erasure.

3.2 Model and Implementation

All experiments are conducted on Qwen2.5-0.5B (24 decoder layers, hidden dimension 896). The current paper-locked comparison set implements five primary erasure methods: linear projection, INLP, LEACE, `gce_ambient`, and `gce_tangent`. LEACE is included as a matched linear baseline in the layer-8 inventory reported below. For `gce_tangent`, we estimate local tangent spaces with PCA-based neighborhoods so that the erasure direction is constrained to a local on-manifold approximation rather than to the full ambient space.

3.3 Evaluation Protocol

Following the evaluation contract in our research brief, all probes are trained from scratch on erased representations to ensure no leakage from the erasure scorers themselves. We report both linear and nonlinear leakage, task-specific utility preservation (answer matching, helpfulness ranking, and profession classification), and manifold-fidelity metrics such as displacement, cosine similarity, and local-geometry diagnostics when available. For the cross-family inventory discussed in the Results section, we use the family-matched leakage evaluator: nonlinear erasers are scored with a fresh

Table 1: Comparison of concept erasure methods by implementation status and role in this work. We distinguish between methods implemented in this repository and those positioned as literature baselines.

Method	Family	Implemented	Replication	Canonical Dataset	Role in Paper
Linear Projection	Linear	Yes	Yes	Bias in Bios	Global baseline
INLP	Linear	Yes	Yes	Bias in Bios	Linear baseline
LEACE	Linear	Yes	Yes	Bias in Bios / Syc.	Matched linear baseline
IGBP	Nonlinear	No	Not yet	Bias in Bios	Prior lineage (proxy only)
`gce_ambient`	Nonlinear	Yes	Yes	Sycophancy	Nonlinear baseline
`gce_tangent`	Nonlinear	Yes	Yes	Sycophancy	Main contribution

Table 2: Representative completed paper-run comparisons at layer 8, $\lambda = 1.0$. Columns T (tangent-constrained) and A (ambient baseline) compare two erasure methods. Utility metrics are task-specific: answer-matching delta for sycophancy, helpfulness delta for safety, and profession control accuracy for Bias in Bios. Leakage and displacement decrease from left to right (lower is better); utility increases from left to right (higher is better). This table reports single completed run points, not exhaustive sweeps; layer-4 analogues exist in the same experimental series. Tangent erasure achieves substantially lower leakage across all three domains but requires higher displacement and produces sharper utility drops.

Domain / Setting	Utility	T leak	A leak	T util	A util	T disp	A disp
Sycophancy L8, $\lambda = 1.0$	Ans. match Δ	0.105	0.231	-20.4pp	-17.0pp	0.728	0.048
Safety L8, $\lambda = 1.0$	Helpful. Δ	0.068	0.252	-0.060	-0.021	2.68	0.20
Bias in Bios L8, $\lambda = 1.0$	Profession acc.	0.062	0.221	0.513	0.536	0.847	0.156

nonlinear MLP probe, while linear erasers are scored with a fresh linear probe. Main-text quantitative claims are restricted to the completed paper suite; broader repository snapshot tables are reported separately in Appendix A.

3.4 Geometric Foundation (Appendix Support)

While the main text focuses on the performance of `gce_tangent`, we also provide geometric diagnostics to ground the mechanism story. These include broad geometry sweeps across multiple datasets and layers, subspace projection sensitivity (Q1), perturbation-based subspace sensitivity tests (Q2), and representation connectivity analyses (Q3). These appendix-scale materials support the low-dimensional and nonlinear-geometry motivation for tangent projection, but they should not be read as a standalone causal proof that off-manifold displacement fully explains the observed utility tradeoffs.

4 Results

This draft separates two evidence tiers. Main-text quantitative claims use only the paper runs that were complete when the manuscript snapshot was assembled: the sycophancy tangent-vs-ambient sweep at layers 4 and 8, the Bias in Bios tangent-vs-ambient sweep at layers 4 and 8, the layer-8 iterative safety sweep, and the completed geometry summaries for gender and sycophancy. Broader repository snapshot tables from completed reports are included in Appendix A for status tracking, but they are not promoted to the main claim set below.

Tables 3 and 4 provide the current matched layer-8 Qwen inventory across the implemented method set. They are intended as a descriptive cross-family comparison: nonlinear erasers are scored with fresh nonlinear probes, while linear erasers are scored with fresh linear probes and their excess-over-chance leakage is floored at zero.

4.1 Sycophancy: Nonlinear Concept Erasure

Table 2 shows the layer-8 comparison at $\lambda = 1.0$. Tangent erasure yields substantially lower leakage (0.105 vs. 0.231), consistent with results across all completed layer/lambda pairs: at layer 4, the best tangent point reaches 0.127 leakage versus 0.168 for ambient.

Table 3: Current matched layer-8 Qwen leakage inventory across the implemented method set. Each entry uses the best `retraction=none` point within the common layer-8 sweep for that method. Following the evaluation contract, nonlinear methods are measured with a fresh nonlinear probe, while linear methods are measured with a fresh linear probe and reported as excess accuracy above chance, floored at zero. Lower is better.

Method	Gender leakage	Sycophancy leakage
`gce_tangent`	0.0621	0.0713
`gce_ambient`	0.2096	0.1033
INLP	0.1979	0.1633
LEACE	0.0000	0.0000
Linear projection	0.1975	0.1633

Table 4: Current matched layer-8 Qwen surgicality inventory across the implemented method set. Entries report `downstream_accuracy_delta`: profession preservation for Bias in Bios and answer-matching delta for sycophancy. Values closer to zero are better.

Method	Gender delta	Sycophancy delta
`gce_tangent`	-0.0250	-0.1800
`gce_ambient`	0.0000	-0.1408
INLP	0.0004	-0.1663
LEACE	-0.0063	-0.1033
Linear projection	0.0004	-0.1663

The lower leakage comes at a cost. Tangent produces larger displacement (0.728 vs. 0.048) and sharper utility loss (-20.4 vs. -17.0 percentage points), patterns that hold across the completed sweep. The evidence therefore supports better concept removal but does not yet establish a frontier advantage in leakage-utility-displacement tradeoffs.

4.2 Safety: Surgicality and Helpfulness

The completed iterative safety sweep at layer 8 shows tangent erasure reducing leakage far more aggressively than ambient (Table 2). At $\lambda = 1.0$, leakage is 0.068 for tangent versus 0.252 for ambient.

Again, lower leakage requires higher cost. Tangent produces larger displacement (2.68 vs. 0.20) and a steeper helpfulness penalty (-0.060 vs. -0.021), indicating a utility-surgicality tradeoff rather than unambiguous improvement.

4.3 Bias in Bios: Diagnostic Baseline

Table 2 shows that tangent reaches substantially lower gender leakage than ambient at layer 8, $\lambda = 1.0$ (0.062 vs. 0.221), mirroring layer-4 results (0.084 vs. 0.213 at best points).

This stronger removal comes with utility and displacement costs. At layer 8, tangent reduces control-task accuracy (0.513 vs. 0.536) and requires larger displacement (0.847 vs. 0.156). The linear anchor therefore supports stronger removal, not yet a cleaner utility-preserving regime.

The same layer-8 Qwen snapshot also provides a descriptive comparison against the implemented linear baselines (Tables 3 and 4). On Bias in Bios, both linear projection and INLP remain near 0.198 leakage with profession accuracy around 0.539, whereas tangent reaches 0.062 leakage with profession accuracy 0.513 and LEACE drives the fresh linear probe to chance with profession accuracy 0.538. On sycophancy, linear projection and INLP both land at 0.163 leakage with answer-matching deltas around -16.6 percentage points, whereas tangent reaches 0.071 leakage with a -18.0 point answer-matching delta and LEACE again drives the fresh linear probe to chance with a smaller utility drop of -10.3 points. We treat these cross-family numbers as descriptive rather than as matched-frontier claims, because the evaluation protocol intentionally uses nonlinear probes for nonlinear erasers and linear probes for linear erasers.

Table 5: Geometry contrast between sycophancy and gender across completed layers 4, 8, and 12. Lower intrinsic dimension and larger local-vs-global angle indicate stronger nonlinearity: a concept whose erasure direction varies significantly across the input space, making a single fixed direction insufficient to capture the full concept. Sycophancy shows substantially higher nonlinearity (dim 34–53, angle 76.9–82.6°) compared to gender (dim 365–461, angle 51.8–55.8°), supporting sycophancy as the primary case for manifold-aware erasure methods. This table demonstrates the geometric contrast but does not establish causality: it does not prove that the observed nonlinearity will be converted into lower displacement or higher utility preservation by any particular erasure method.

Concept	Local intrinsic dimension	Mean local/global angle
Sycophancy	34–53	76.9–82.6°
Gender	365–461	51.8–55.8°

4.4 Geometric Evidence and Off-Manifold Displacement

Table 5 demonstrates that sycophancy is much more nonlinear than gender: its intrinsic dimension (34–53 across layers 4, 8, 12) is far lower than gender's (365–461), and its local-vs-global angle is much larger (76.9–82.6° vs. 51.8–55.8°). Pairwise scorer agreement remains high in the completed runs, suggesting coherence in the local direction field despite weak global summary.

This geometry contrast validates sycophancy as the primary case for manifold-aware erasure methods. However, the completed paper runs do not yet prove that this geometric advantage translates into lower-displacement or higher-utility erasure points on any frontier.

Appendix-Supporting Evidence. Detailed geometric analyses supporting these claims are provided in the Appendix, including:

- **Subspace Sufficiency (Q1):** Evidence that task-specific computation is concentrated in a low-dimensional linear subspace.
- **Perturbation Sensitivity (Q2):** Comparison of perturbations in high-variance and orthogonal subspaces, which supports low-dimensional sensitivity concentration but does not by itself prove curved-manifold causality.
- **Representation Connectivity (Q3):** Analysis of the connected, approximately convex topology of the representation manifold.
- **Multi-Layer Accumulation:** Results on how projection error accumulates across the network.

These appendix-scale geometry analyses remain important support for the mechanism story, but the main-text quantitative claims should currently be limited to the completed paper runs summarized above.

Broader Repository Snapshot. The broader iterative reports recorded in Appendix A are directionally consistent with the main-text picture. In both the sycophancy and safety repository snapshots (Tables 6 and 7), deeper-layer iterative tangent runs continue to reduce leakage more than ambient baselines. We include those tables to document the current state of the project and to motivate the next matched-leakage analyses, but we do not rely on them for the central paper claims.

5 Discussion

The completed paper suite already narrows the central claim. In every finished comparison, the tangent-based method removes more measured concept signal than its ambient counterpart. That pattern appears in the sycophancy sweeps, in the completed layer-8 safety sweep, and in the Bias in Bios diagnostic. What the current paper snapshot does not yet show is the cleaner second half of the original thesis in which tangent projection also achieves lower displacement or uniformly better utility retention. In the completed runs, tangent reaches lower leakage by taking larger steps, and its utility metrics are either slightly worse or mixed.

The geometry diagnostics still matter because they explain why sycophancy is the harder case. Sycophancy shows local-vs-global angles near 80°, while gender remains near 53°, so a single global erasure direction is a much worse summary for the primary nonlinear concept. That is consistent with tangent-based methods improving removal in both concepts while leaving sycophancy as the clearly more nonlinear case, but the finished results currently frame tangent as a stronger remover rather than a dominantly better Pareto point.

The appendix snapshot tables slightly sharpen the near-term agenda. In the broader repository reports (Tables 6 and 7), the same removal advantage continues at layers 16 and 23 for both sycophancy and safety, although the utility metrics differ by domain and report. We do not elevate those rows to headline claims because they were not part of the paper-locked evidence set, but they strengthen the case for spending the next round of effort on matched-leakage comparisons and deeper-layer replications rather than on searching for a different primary mechanism.

Two implications follow. First, the paper should presently avoid claiming that off-manifold control is already proven to improve the full leakage-utility frontier; the matched-leakage analysis and broader replications are still needed for that claim. Second, the current evidence suggests that the next important question is not whether tangent projection changes the erasure direction at all, but under which layers and operating points that removal advantage can be converted into a cleaner utility tradeoff.

6 Limitations

The current manuscript deliberately limits its main-text quantitative claims to the paper runs that were complete when this draft was assembled. That keeps the claims auditable, but it also means the strongest version of the utility-preservation story is not yet established in the locked evidence set. The deeper iterative sycophancy and safety reports are directionally consistent with the main narrative and are included in Appendix A, yet the paper's central claims still rely on the smaller completed suite.

A second limitation is measurement scope. All experiments use Qwen2.5-0.5B, and utility is measured through task-specific proxies such as answer matching, helpfulness ranking, profession classification, and dataset-source retention. These metrics are appropriate for surgicality, but they do not yet establish broad capability preservation across unrelated tasks.

Finally, the geometry diagnostics support low-dimensional structure, concept nonlinearity, and connectedness, but they do not by themselves prove that curved-manifold effects are the sole causal mechanism behind every leakage–utility tradeoff. The current paper therefore treats geometry as strong supporting evidence for tangent projection rather than as a complete causal proof.

7 Conclusion

The completed paper runs already support a narrower, evidence-backed conclusion. Sycophancy is geometrically more nonlinear than Bias in Bios, and tangent-based erasure consistently achieves lower measured concept leakage than the completed ambient baselines on sycophancy, safety, and gender. That makes manifold-aware erasure a supported hypothesis for stronger concept removal in nonlinear settings.

At the same time, the current draft should stop short of claiming that tangent erasure already dominates the utility-fidelity tradeoff. In the finished runs, the lower-leakage tangent points typically require larger displacement and often incur equal or larger utility loss than ambient baselines. Appendix snapshot tables are directionally consistent with the same removal advantage at deeper layers, but the paper is strongest if it uses them as status evidence rather than as the basis for stronger causal or Pareto-frontier claims. The manuscript therefore lands best as support for the geometry contrast and for tangent's removal advantage, while reserving the broader utility-preservation claim for matched-leakage analyses and future replications.

A Repository Snapshot Tables

This appendix records broader repository snapshot tables that are directionally consistent with the main text but were not part of the locked paper-evidence set when the quantitative claims in Section 4

Table 6: Repository snapshot from the iterative sycophancy report at $\lambda = 1.0$ after 7 rounds, across layers 8, 16, and 23. Columns T (tangent-constrained) and A (ambient baseline) compare erasure methods. Utility is answer-matching delta relative to clean representations. Leakage and displacement decrease from left to right (lower is better); utility increases from left to right (higher is better). This snapshot table is not a main-text claim anchor and should be read only as evidence of directional consistency with the layer-8 paper-locked row (Table 2). Deeper-layer iterative tangent runs continue to reduce leakage more than ambient baselines across this exploratory sweep.

Layer	T leak	A leak	T util	A util	T disp	A disp
8	0.063	0.213	-20.7pp	-0.5pp	2.88	0.62
16	0.153	0.400	+0.0pp	+0.0pp	3.79	0.27
23	0.138	0.361	-0.1pp	-0.2pp	44.50	3.11

Table 7: Repository snapshot from the iterative safety report at $\lambda = 1.0$ after 7 rounds, across layers 8, 16, and 23. Columns T (tangent-constrained) and A (ambient baseline) compare erasure methods. Utility is pairwise helpfulness delta. Leakage and displacement decrease from left to right (lower is better); utility increases from left to right (higher is better). This snapshot table is not a main-text claim anchor and should be read only as evidence of directional consistency with the layer-8 paper-locked row (Table 2). Deeper-layer iterative tangent runs continue to reduce leakage more than ambient baselines across this exploratory sweep.

Layer	T leak	A leak	T util	A util	T disp	A disp
8	0.0725	0.2583	-0.052	-0.023	2.70	0.18
16	0.1292	0.3517	-0.092	+0.003	4.37	0.60
23	0.1483	0.3404	-0.062	-0.152	38.32	5.30

were written. We include them to document the current state of the project and to guide the next round of matched-leakage and deeper-layer analyses. The central paper claims continue to rely only on the completed paper suite summarized in the main text.

A.1 Iterative Sycophancy Snapshot

A.2 Iterative Safety Snapshot

Geometry Note. The broader geometry report continues to support three useful points: early and mid-layer representations are effectively low-dimensional, on-manifold directions carry much of the model's sensitivity in these perturbation tests, and interpolation remains consistent with a connected representation space. At the same time, the Q2 perturbation study is best read as evidence of sensitivity concentration in a high-variance subspace rather than as a standalone proof of curved-manifold causality.

Table 8: Geometry evidence map organizing the appendix diagnostics into four research blocks. Each block addresses a distinct aspect of manifold structure: Q1 examines whether task-relevant computation concentrates in a low-dimensional linear subspace, Q2 tests whether perturbations on the manifold are more consequential than off-manifold perturbations, Q3 validates whether the manifold is connected across domains, and multi-layer accumulation tracks error compounding through the network. These diagnostics support the mechanism discussion and motivate the need for manifold-aware erasure methods but do not by themselves establish causality.

Experiment Block	Research Question	Primary Evidence
Q1: Rank-retention	Subspace concentration	Top-1 agreement under intrinsic projection
Q2: Perturbation sensitivity	High-variance vs. orthogonal directions	KL and top-1 under on/off-subspace noise
Q3: Domain connectivity	Shared vs. domain manifolds	Cross-domain interpolation smoothness
Multi-layer accumulation	Error compounding across the network	Simultaneous intrinsic projection across layers

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